

AUTOMATED WALL PAINTING MACHINE

Bill of Materials — Working Prototype

Final Year Project | B.E. Mechanical Engineering | Don Bosco College of Engineering, Goa University | 2022–23

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Bill of Materials

Sr. No.	Component / Part Name	Specification & Description	Qty	Unit (INR)	Total (INR)
1	PMDC Motor	350W, 24V DC, high-torque permanent magnet DC motor — primary vertical drive for lifting Frame 2 and paint dispensing system	1	3,200	3,200
2	Geared Box Motor	12V DC, gear ratio 1:150 — drives lateral track carriage, 75mm step per actuation	1	850	850
3	Mild Steel Sections (Frame)	IS 2062 mild steel square hollow sections (20mm × 20mm, T=2mm) — Frame 1, Frame 2, and base structural members; stress-validated via ANSYS FEA	lot	—	4,800
4	Base Plate & Casing	MS sheet metal base housing enclosing motor, Arduino, power supply, and wiring harness; dismountable for transport	1	1,100	1,100
5	Lateral Track	2-metre aluminium extrusion guide track with carriage rails for sideways (lateral) movement of the entire frame assembly	1	1,400	1,400
6	Rope-Pulley System	Nylon rope (6mm dia.) + 4 grooved pulleys — converts PMDC motor rotational output to vertical linear motion of paint system	1 set	620	620
7	Track Guide Wheels & Bearings	Grooved nylon wheels + ball bearings for smooth low-friction lateral carriage movement on aluminium track	4	180	720
8	Fasteners & Hardware	M6 and M8 hex bolts, nylock nuts, flat washers, locknuts — complete frame assembly; all joints are demountable	lot	350	350
9	HVLP Spray Gun	High Volume Low Pressure spray gun, gravity-feed cup, 1.4mm nozzle, max 100 PSI operating pressure — main paint application device	1	1,800	1,800
10	Linear Actuator (100N)	12V DC, 100N output force, 30mm stroke — trigger mechanism for HVLP gun (final iteration after servo motor and solenoid valve both failed due to insufficient force)	1	1,200	1,200
11	3D-Printed Mounting Bracket	Custom FDM-printed PLA bracket designed in SolidWorks to mount HVLP gun securely to frame carriage — replaced mid-testing after bracket slippage in final trial	1	220	220
12	Air Hose & Fittings	6mm polyurethane air hose + push-fit quick-release couplings connecting external compressor to HVLP gun	1 set	480	480

13	Arduino Uno R3	ATmega328P microcontroller — central control unit running the spray-move-detect-shift state machine logic; programmed in C (Arduino IDE)	1	550	550
14	HC-SR04 Ultrasonic Sensors	Range 2cm–400cm, 5V supply — two units for obstacle detection (windows, switchboards) during vertical spray cycle; pauses spray gun on detection	2	120	240
15	Limit Switches (Mechanical)	SPDT mechanical micro-switches — top and bottom position detection for vertical drive end-stop control	2	85	170
16	L298N Motor Driver Module	Dual H-bridge motor driver, 2A per channel — independent PWM speed control for PMDC motor and geared box motor from Arduino digital outputs	1	280	280
17	5V Relay Module	Single-channel 5V relay — switches linear actuator (100N) ON/OFF via Arduino digital signal; provides electrical isolation	1	120	120
18	Power Supply Unit	24V DC, 5A SMPS regulated supply for PMDC motor; 12V adapter for Arduino, relay, and sensor circuits	1	950	950
19	Wiring & PCB Stripboard	Connecting wires (22AWG), stripboard, terminal blocks, heat-shrink tubing — clean wiring harness inside base casing	lot	380	380
20	Switches & Safety Controls	Main power ON/OFF toggle + emergency stop push-button mounted on base casing for operator safety	2	95	190
21	Water-Based Emulsion Paint	Standard white emulsion, 1L, medium viscosity — used for prototype validation testing at 190mm gun-to-wall distance, 100 PSI	1	180	180
22	External Air Compressor	1HP single-stage air compressor, 100 PSI rated — used during testing only; not physically part of the machine (external equipment)	1 (ext.)	—	—
23	Miscellaneous Consumables	Protective paint for steel frame (rust prevention), adhesives, cable ties, fastener spares, prototyping material for bracket iterations	lot	457	457
				TOTAL	INR 27,857

Cost Summary

Cost Summary	Amount	
Mechanical Structure (Items 1–8)		INR 13,040
Paint Dispensing System (Items 9–12)		INR 3,700
Electronics & Control (Items 13–20)		INR 2,660
Miscellaneous & Consumables (Items 21–23)		INR 637
TOTAL PROTOTYPE BUILD COST		INR 27,857
Commercial alternative (MYRO robot, equivalent)		INR 15,00,000+
Cost saving vs. commercial alternative		~98.1%

Important Note — Working Prototype Declaration

This Bill of Materials documents the components used in the construction of a **working prototype** of the Automated Wall Painting Machine, developed as a Final Year Engineering Project at Don Bosco College of Engineering, Goa University (2022–23). The prototype was designed, fabricated, assembled, and validated by a team of five undergraduate students within the constraints of an academic project budget of INR 27,857.

This prototype **is not a commercially ready or fully productised system**. It was built to demonstrate proof-of-concept for automated wall painting using a rope-pulley vertical drive system, a track-based lateral movement mechanism, ultrasonic obstacle detection, and Arduino-based closed-loop control logic. The machine successfully completed 12 test runs and painted a 1.5m × 0.6m test board in approximately 238 seconds under controlled conditions.

The following limitations were identified during prototype validation and represent known engineering gaps that would require resolution before any production or commercial deployment:

- Paint dispensing bracket slipped mid-run during the final validation trial — the mounting clamp mechanism requires redesign for production use
- HVLP spray gun requires an external air compressor — portability constraint not resolved at prototype stage
- Machine currently paints vertical strips only (bottom-to-top); no horizontal or diagonal capability
- Obstacle detection is basic ultrasonic threshold-based logic — computer vision and adaptive path planning are identified as future development areas

The BOM reflects actual single-unit procurement costs from local Indian suppliers (Goa, 2023). Production-scale costs and component specifications would differ significantly in a commercial product development context. All components listed were physically integrated into the prototype and verified during assembly and testing phases.

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